



Information Integration project

Andrea Lanzarone
Claudia Cornacchia





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2

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Phase 1 – Data Gathering

Scenario of interest : Housing rental market in Rome, comparing

Short-term rentals (Airbnb)

Long-term rentals

Considering also:

- Neighborhood population
- Proximity to public transport
- Tourist attractions.



Dataset

Short term rentals



- Source: InsideAirbnb.
- Contains: name, price, room type, accommodates, review score, latitude and longitude ...



- Source: Kaggle “Italian Airbnb Dataset” filtering only Rome.
- Contains: name, price, property type, number of allowed guests, rating score, coordinates, city ...



Long term rentals

- Source: Kaggle “Italy house prices”
- Contains: price, square meters, address ...



Point of interest and metro

- Source: OpenStreetMap database
- Contains: name, type, coordinates



Population

- Source: official Rome geoportal
- Contains: municipio, quartiere, number of residents



Data Heterogeneity

Physical Heterogeneity

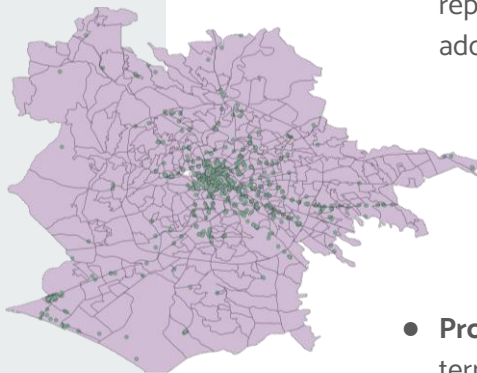
(Data are stored in different formats)



- **Prices** were stored as strings with a dollar symbol in Inside Airbnb but as integers in the Kaggle dataset.
- **Review scores** were integers in the range 0 – 500 in Inside Airbnb, while in the other were floats between 0.00 and 5.00.



Solved at the **Wrapper** level



Conceptual Heterogeneity

(Data represent different aspects of the same phenomena)



- Each dataset used different **location** representations (latitude/longitude or addresses).
 - We used a GeoJSON file containing the polygon boundaries of all neighborhoods in Rome and performed spatial joins to associate each record to the corresponding neighborhood (procedural mapping).
- **Property size** was expressed in m² for long-term rentals and number of allowed guests for short-term rentals.



Solved at the **Mapping** level

Set of tasks



1. **Tourist attraction proximity:** Search for highly rated Airbnb close to many tourist attractions.



2. Identify neighborhoods in Rome experiencing **over-tourism**. The task returns neighborhood with fewer than 1,000 residents, at least three distinct Airbnb listings, and long-term rental prices exceeding 20 € per square meter.

3. **Public transport and affordability:** Search for affordable rentals near at least two metro stations



4. **Break Even days:** For each Airbnb listing compute the number of rental days per month required to match the average monthly long term rental income in the same neighborhood and for the same property size (S,M,L).

Phase 2 – Modelling

Information integration system $J = \langle G, M, \emptyset \rangle$

- **Global schema** $\rightarrow$ logical theory over A_G

Including Equality Generating Dependency (EGD), a first-order formula of the form:

$$\forall \bar{x}. \varphi(\bar{x}) \rightarrow (x_i = x_j)$$

where φ is a conjunction of relational atoms.

- **Source schema** $\rightarrow$ empty logical theory over A_S .

- **Mapping** $\rightarrow$ a set of GAV mapping assertions:

Pair $\langle q_S, g \rangle$ where:

- $q_S = \{\bar{x} \mid \varphi_S(\bar{x})\}$ is a FOL query over the source schema
- g is an atomic query for the global schema.

Phase 2 – Modelling

IIS Semantics:

Given a database D that satisfies S ($D \models S$), the Information Integration **Semantics** of J with respect to D is the **set of databases** C such that:

- $C \models G$
- $q_S^D \subseteq q_G^C$ for each $\langle q_S^D, q_G^C \rangle \in M$

Classification of our IIS:

- **Scope of Integration** $\rightarrow$ Domain-Based IIS
- **Mapping** $\rightarrow$ Declarative
- **Result of the integration** $\rightarrow$ Databases
- **Representation of the Result** $\rightarrow$ Materialization



Source Schema

$A_S = \{$

S_PointOfInterest(id, nome, turistiche, storiche, metro, quartiere)

S_AirbnbInside(id, name, price, room_type, accommodates, review_scores_rating, quartiere)

S_AirbnbKaggle(Listings_id, Name, Price, Property_type, Maximum_allowed_guests,
Rating_score, quartiere)

S_Rental(id, prezzo, m2, via, quartiere, price_per_sqm)

S_Population(quartiere, municipio, num_residenti)

$\}$

$S = \emptyset$

Global Schema

$A_G = \{$

G_NeighborhoodInterest(id, name, neighborhood, category)

G_ShortTerms (id, name, price_daily, rating, neighborhood, size)

G_LongTerms (id, price_monthly, price_per_sqm, size, neighborhood, m2)

G_NeighborhoodPopulation (neighborhood, num_residents)

$\}$

Axioms (EGD):

- $\forall \text{id, cat, } n_1, n_2, q_1, q_2. \text{G_NeighborhoodInterest}(\text{id}, n_1, q_1, \text{cat}) \wedge \text{G_NeighborhoodInterest}(\text{id}, n_2, q_2, \text{cat}) \rightarrow (n_1 = n_2 \wedge q_1 = q_2)$
- $\forall \text{id, } n_1, p_1, r_1, \text{qrt}_1, s_1, n_2, p_2, r_2, \text{qrt}_2, s_2. \text{G_ShortTerms}(\text{id}, n_1, p_1, r_1, \text{qrt}_1, s_1) \wedge \text{G_ShortTerms}(\text{id}, n_2, p_2, r_2, \text{qrt}_2, s_2) \rightarrow (n_1 = n_2 \wedge p_1 = p_2 \wedge r_1 = r_2 \wedge \text{qrt}_1 = \text{qrt}_2 \wedge s_1 = s_2)$
- $\forall \text{id, } pm_1, pps_1, s_1, \text{qrt}_1, m_1, pm_2, pps_2, s_2, \text{qrt}_2, m_2. \text{G_LongTerms}(\text{id}, pm_1, pps_1, s_1, \text{qrt}_1, m_1) \wedge \text{G_LongTerms}(\text{id}, pm_2, pps_2, s_2, \text{qrt}_2, m_2) \rightarrow (pm_1 = pm_2 \wedge pps_1 = pps_2 \wedge s_1 = s_2 \wedge \text{qrt}_1 = \text{qrt}_2 \wedge m_1 = m_2)$
- $\forall \text{qrt, } nr_1, nr_2. \text{G_NeighborhoodPopulation}(\text{qrt}, nr_1) \wedge \text{G_NeighborhoodPopulation}(\text{qrt}, nr_2) \rightarrow (nr_1 = nr_2)$

GAV Mapping

G_NeighborhoodInterest

$$\begin{aligned} & \{(id, nome, cat, quartiere) \mid \exists s, m. \text{S_PointOfInterest}(id, nome, 'museum', s, m, quartiere) \wedge cat = 'tourist attraction'\} \\ & \quad \cup \\ & \{(id, nome, cat, quartiere) \mid \exists s, m. \text{S_PointOfInterest}(id, nome, 'attraction', s, m, quartiere) \wedge cat = 'tourist attraction'\} \\ & \quad \cup \\ & \{(id, nome, cat, quartiere) \mid \exists t, m. \text{S_PointOfInterest}(id, nome, t, 'archaeological_site', m, quartiere) \wedge cat = 'tourist attraction'\} \\ & \quad \cup \\ & \{(id, nome, cat, quartiere) \mid \exists t, m. \text{S_PointOfInterest}(id, nome, t, 'ruins', m, quartiere) \wedge cat = 'tourist attraction'\} \\ & \quad \cup \\ & \{(id, nome, cat, quartiere) \mid \exists t, m. \text{S_PointOfInterest}(id, nome, t, 'monuments', m, quartiere) \wedge cat = 'tourist attraction'\} \\ & \quad \cup \\ & \{(id, nome, cat, quartiere) \mid \exists t, s. \text{S_PointOfInterest}(id, nome, t, s, 'station', quartiere) \wedge cat = 'metro'\} \\ & , \{(id, name, category, neighborhood) \mid \text{G_NeighborhoodInterest}(id, name, category, neighborhood) \} \} \end{aligned}$$

GAV Mapping

G_ShortTerms

$$\begin{aligned} & \{ \{ (id, n, p, size, r, qrt) \mid \exists acc. \text{S_AirbnbInside}(id, n, p, 'Entire home/apt', acc, r, qrt) \wedge acc \leq 2 \wedge size = 'S' \} \\ & \quad \cup \\ & \{ (id, n, p, size, r, qrt) \mid \exists acc. \text{S_AirbnbInside}(id, n, p, 'Entire home/apt', acc, r, qrt) \wedge acc > 2 \wedge acc \leq 5 \wedge size = 'M' \} \\ & \quad \cup \\ & \{ (id, n, p, size, r, qrt) \mid \exists acc. \text{S_AirbnbInside}(id, n, p, 'Entire home/apt', acc, r, qrt) \wedge acc > 5 \wedge size = 'L' \} \\ & \quad \cup \\ & \{ (id, n, p, size, r, qrt) \mid \exists max_g. \text{S_AirbnbKaggle}(id, n, p, 'Entire home', max_g, r, qrt) \wedge max_g \leq 2 \wedge size = 'S' \} \\ & \quad \cup \\ & \{ (id, n, p, size, r, qrt) \mid \exists max_g. \text{S_AirbnbKaggle}(id, n, p, 'Entire home', max_g, r, qrt) \wedge max_g > 2 \wedge max_g \leq 5 \wedge size = 'M' \} \\ & \quad \cup \\ & \{ (id, n, p, size, r, qrt) \mid \exists max_g. \text{S_AirbnbKaggle}(id, n, p, 'Entire home', max_g, r, qrt) \wedge max_g > 5 \wedge size = 'L' \} \\ & , \{ (id, name, price_daily, size, rating, neighborhood) \mid \text{G_ShortTerms}(id, name, price_daily, size, rating, neighborhood) \} \} \end{aligned}$$



GAV Mapping

G_LongTerms

$$\begin{aligned} & \langle \{ (id, prezzo, price_per_sqm, size, quartiere, m2) \mid \exists \text{ via} . \text{S_Rental}(id, prezzo, m2, \text{via}, quartiere, price_per_sqm) \wedge m2 \leq 50 \wedge size = 'S' \} \\ & \quad \cup \\ & \{ (id, prezzo, price_per_sqm, size, quartiere, m2) \mid \exists \text{ via} . \text{S_Rental}(id, prezzo, m2, \text{via}, quartiere, price_per_sqm) \wedge m2 > 50 \wedge m2 \\ & \leq 100 \wedge size = 'M' \} \\ & \quad \cup \\ & \{ (id, prezzo, price_per_sqm, size, quartiere, m2) \mid \exists \text{ via} . \text{S_Rental}(id, prezzo, m2, \text{via}, quartiere, price_per_sqm) \wedge m2 > 100 \wedge size = 'L' \} \\ & \quad , \{ (id, price_monthly, price_per_sqm, size, neighborhood, m2) \mid \text{G_LongTerms}(id, price_monthly, price_per_sqm, size, neighborhood, m2) \} \rangle \end{aligned}$$



GAV Mapping

G_NeighborhoodPopulation

$\langle \{ (\text{quartiere}, \text{num_residenti}) \mid \exists \text{ municipio . } \text{S_Population}(\text{quartiere}, \text{municipio}, \text{num_residenti}) \}$
 $, \{ (\text{neighborhood}, \text{num_residents}) \mid \text{G_NeighborhoodPopulation}(\text{neighborhood}, \text{num_residents}) \} \rangle$

Query answering decision problem

Once defined the IIS, the goal is to be able to query the reconciled view of the data to obtain certain answers.

Certain answers for q w.r.t. J and D are the set:

$$\text{cert}(q, J, D) = \{\bar{a} \mid \bar{a} \in q(B) \text{ for each } B \in \text{sem}^D(J)\}$$

OWA $\rightarrow$ infinite possible models

Theorem: Assume a database D that satisfies S and such that $\text{sem}^J(D) \neq \emptyset$ (consistent). Given a **UCQ** $q(\bar{x})$ and a tuple $\bar{c}$, $\bar{c} \in \text{cert}(q, J, D)$ if and only if $\bar{c} \in q(\mathbf{M}(D))$

So we used the materialization approach :

- Materialize the Retrieved Global Database
- Check whether the **Retrieved Global Database satisfies the constraints**
- Use the Retrieved Global Database to answer UCQs



Query 1 – Tourist attraction proximity

Find the Airbnb listings that have a review score above 4.8 and are located in neighborhoods that contain at least 3 distinct tourist attractions.

$q = \{ (name, price_daily, rating, neighborhood) \mid$
 $\exists id, size, id_{i1}, name_{i1}, id_{i2}, name_{i2}, id_{i3}, name_{i3} .$
 $G_ShortTerms(id, name, price_daily, rating, neighborhood, size) \wedge$
 $rating > 4.8 \wedge$
 $G_NeighborhoodInterest(id_{i3}, name_{i3}, neighborhood, 'tourist attraction') \wedge$
 $G_NeighborhoodInterest(id_{i2}, name_{i2}, neighborhood, 'tourist attraction') \wedge$
 $G_NeighborhoodInterest(id_{i1}, name_{i1}, neighborhood, 'tourist attraction') \wedge$
 $id_{i1} \neq id_{i2} \wedge id_{i1} \neq id_{i3} \wedge id_{i2} \neq id_{i3} \}$

Query 2 - Over tourism

Find neighborhood with fewer than 1,000 residents, at least three distinct Airbnb listings, and long-term rental prices greater than €20 per square meter

$$q = \{ (id_{lt}, neighborhood, num_residents, price_per_sqm) \mid$$
$$\begin{aligned} & \exists id_1, n_1, p_{d1}, s_1, r_1, id_2, n_2, p_{d2}, s_2, r_2, id_3, n_3, p_{d3}, s_3, r_3, price_monthly, s_{lt}, m2 . \\ & G_NeighborhoodPopulation(neighborhood, num_residents) \wedge num_residents < 1000 \wedge \\ & G_ShortTerms(id_1, n_1, p_{d1}, r_1, neighborhood, s_1) \wedge \\ & G_ShortTerms(id_2, n_2, p_{d2}, r_2, neighborhood, s_2) \wedge \\ & G_ShortTerms(id_3, n_3, p_{d3}, r_3, neighborhood, s_3) \wedge id_1 \neq id_2 \wedge id_1 \neq id_3 \wedge id_2 \neq id_3 \wedge \\ & G_LongTerms(id_{lt}, price_monthly, price_per_sqm, s_{lt}, neighborhood, m2) \wedge \\ & price_per_sqm > 20 \} \end{aligned}$$



Query 3 - Public transport and affordability

Find long-term rental that are in a neighborhood with at least 2 metro stations and with a cost per month lower than 600.

$$q = \{ (id, neighbourhood, price_monthly, m2) \mid$$
$$\exists size, price_per_sqm, id_{m1}, name_{m1}, id_{m2}, name_{m2} .$$
$$G_LongTerms(id, price_monthly, price_per_sqm, size, neighborhood, m2) \wedge$$
$$G_NeighborhoodInterest(id_{m1}, name_{m1}, neighborhood, 'metro') \wedge$$
$$G_NeighborhoodInterest(id_{m2}, name_{m2}, neighborhood, 'metro') \wedge$$
$$id_{m1} \neq id_{m2} \wedge price_monthly < 600 \}$$

Query 4 - Break Even days

Aggregation cannot be formulated as a Union of Conjunctive Queries so there is no guarantee to retrieve Certain Answers



Task → Break Even Days = **Average** monthly long-term rent for that size in that neighborhood/ Airbnb nightly price.

Alternative CQ query:

Returns for each short term rental its daily price compared with the monthly price of the long term rental (in the same neighborhood and with the same size).

$$q = \{ (name, price_daily, size, neighborhood, price_monthly) \mid \\ \exists id_{st}, rating, id_{lt}, price_sqm, m2 . \\ G_LongTerms(id_{lt}, price_monthly, price_sqm, size, neighborhood, m2) \wedge \\ G_ShortTerms(id_{st}, name, price_daily, rating, neighborhood, size) \}$$

Phase 3 – Implementation

ETL Tool:

